

THE GRAND FRACTION BAKERY

MULTIPLYING MADE EASY!

Multiplying a Fraction by a Whole Number

Grade 4 Mathematics Lesson Guide

What happens when you have many of the same fraction parts?

Let's discover how combining fraction groups is just a speedy multiplication trick!

 Welcome Little Chefs and Mathematicians! 

Welcome to the Cookie Party!

Imagine baking cookies with your best friends on a sunny Saturday afternoon!

Let's meet three hungry friends: **Anya**, **Rahul**, and **Zayn**. They are sitting at a kitchen table, and their teacher places a cookie sheet down. But instead of whole cookies, they are looking at cut-up pieces!

- **Anya** has $\frac{1}{4}$ (one-quarter) of a giant chocolate cookie.
- **Rahul** has $\frac{1}{4}$ of a giant chocolate cookie.
- **Zayn** has $\frac{1}{4}$ of a giant chocolate cookie.

They want to put all their cookie plates together onto one main serving tray to see how much total cookie they have. How can we write this as a math equation?

Let's Guess:

We could add them up one by one like this: $\frac{1}{4} + \frac{1}{4} + \frac{1}{4}$. But is there a faster way to count matching groups using multiplication? Let's check it out!

+ Adding the Same Thing Over and Over

Do you remember what multiplication really means? It is just a shortcut for adding the exact same number over and over again!

For example, if you have 3 boxes, and each box contains 2 apples, you don't need to count $2 + 2 + 2$. You can just quickly say:

$$3 \text{ groups of } 2 = 3 \times 2 = 6 \text{ apples!}$$

Fractions work exactly the same way! Let's look back at our cookie friends. We have 3 kids, and each kid has a slice of size $\frac{1}{4}$. That means we have:

$$3 \text{ groups of } \frac{1}{4} = 3 \times \frac{1}{4}$$



Big Math Discovery:

When we multiply a fraction by a whole number, we are simply finding out how many total pieces we have when we collect matching fraction groups together!

The Secret Rule of Fraction Multiplication

Now, let's learn the secret trick to solving $3 \times \frac{1}{4}$ in a blink of an eye. It's so easy that you will feel like a math magician!

Step 1: Multiply the Top Number Only!

Take your whole number (3) and multiply it by the top number of your fraction (1).

$$3 \times 1 = 3$$

This tells us we have collected 3 slices in total!

Step 2: Keep the Bottom Number the Same!

Do NOT change the bottom number. Leave it as 4. Why? Because the size of the slices hasn't changed at all! They are still quarter-sized slices.



Bringing it together:

$$3 \times \frac{1}{4} = \frac{3}{4}!$$

Together, our three friends have three-quarters ($\frac{3}{4}$) of a whole cookie!

Why Doesn't the Bottom Number Change?

Let's make sure we understand why the bottom number stays asleep during multiplication!

Imagine you have 4 plates. On each plate, you put 1 slice of pizza. Each slice is a standard $\frac{1}{8}$ size piece of a large pizza wheel.

If you gather all 4 slices and put them into a cardboard box, you have 4 slices. But are those slices suddenly smaller or larger? No! They are still $\frac{1}{8}$ size slices.

The Label Rule:

The bottom number is just a label that tells us the name or size of the slice. If you multiply 4×1 apple, you get 4 apples, not 4 giant mega-apples. In the same way, $4 \times \frac{1}{8}$ equals $\frac{4}{8}$!

⚡ Step-by-Step Math Magic

Let's look at another clean problem to write down in our math notebooks. Let's solve:

$$5 \times \frac{2}{3}$$

This means we are collecting 5 separate groups, and every single group contains $\frac{2}{3}$ of a sheet.

Action Plan	How We Calculate It
Multiply Top Numbers	$5 \times 2 = 10$ (This goes on top)
Keep Bottom Same	3 stays at the bottom
Final Answer	$\frac{10}{3}$

Wow! Our answer is $\frac{10}{3}$. Because the top number is bigger than the bottom number, it means we have collected enough slices to make multiple whole objects! This is called an **improper fraction**.



Story Problem: Mango Juice Cartons

Let's use our new math powers to solve a real-world story problem about an after-school picnic party!

Meera is setting up juice cups for her friends. She opens 6 small boxes of sweet mango juice. Each box is not quite full—each one contains exactly $\frac{3}{10}$ of a liter of juice.

How much juice does Meera have in total if she combines all 6 boxes together?



Let's Solve Step-by-Step:

- **Step 1:** Write down the expression: $6 \times \frac{3}{10}$
- **Step 2:** Multiply the whole number by the top! $6 \times 3 = 18$
- **Step 3:** Keep the bottom label identical! The denominator stays as 10.
- **Step 4:** Bring it together: The total amount of juice is $\frac{18}{10}$ liters!

Perfect! Meera has exactly $\frac{18}{10}$ liters of delicious mango juice ready to pour!

Practice Problem: Birthday Cake Ribbons

Let's help a baker decorate some beautiful celebration cakes!

A baker needs to put decorative green ribbons around 4 matching birthday cakes. Each cake requires exactly $\frac{2}{5}$ of a meter of ribbon wrapped around its base.

What is the total length of green ribbon the baker needs to buy from the market?

Your Turn to Solve:

1. What is the math equation? $4 \times \frac{2}{5}$
2. Multiply the big whole number by the top numerator: $4 \times 2 = \underline{\hspace{2cm}}$
3. What number stays on the bottom? The number is $\underline{\hspace{2cm}}$
4. Write your final fraction answer: $\underline{\hspace{2cm}}$

☒ **Answer Check:** Did you get $\frac{8}{5}$ meters? Yes! You are getting super good at this!

Reading Clues: The Magic Word "OF"

Sometimes, math word problems don't show a clear multiplication sign ($\times$). Instead, they hide the math operation behind a tiny secret code word.

That secret code word is **"OF"**!



The Word Clue:

Whenever a math problem asks you to find a fraction "of" a whole number group, it means you need to use **multiplication**!

Let's see an example: *"Rohan has a bag of 12 marbles. $\frac{1}{4}$ of his marbles are blue. How many blue marbles does he have?"*

This translates directly into: $\frac{1}{4} \times 12$

Let's do our magic: $1 \times 12 = 12$ on top, and 4 on the bottom, giving us $\frac{12}{4}$. Since 12 divided by 4 equals 3, Rohan has 3 blue marbles!

★ You Are a Fraction Multiplication Hero!

Incredible job completing this guide! Let's lock in our memory checklist:

- ★ **Top Multiplication:** Only multiply the whole number by the top numerator number.
- ★ **Bottom Shield:** Never change or multiply the bottom denominator number.
- ★ **The "Of" Clue:** Remember that finding a fraction "of" something means multiplying!

The Final Master Challenge!

Can you solve this final riddle? What is $7 \times \frac{3}{8}$?

(Think: Multiply 7 by 3 for the top, keep 8 on the bottom. Write your answer down and show it proudly to your teacher!)

FRACTION MULTIPLICATION PRACTICE

Grade 4 Practice Sheet (No Answers - Show Your Work!)

1. Solve: $4 \times \frac{1}{5} =$

2. Solve: $3 \times \frac{2}{7} =$

3. Solve: $6 \times \frac{1}{8} =$

4. Solve: $5 \times \frac{3}{4} =$

5. Solve: $2 \times \frac{4}{9} =$

FRACTION MULTIPLICATION PRACTICE

Word Problems & Master Challenges

6. Find $\frac{1}{3}$ of 15.

7. Find $\frac{2}{5}$ of 10.

8. Solve: $8 \times \frac{2}{3} =$

9. Solve: $7 \times \frac{1}{6} =$

10. Solve: $9 \times \frac{3}{10} =$

FRACTION MULTIPLICATION PRACTICE

Real-World Application Problems

11. A recipe for one tray of brownies requires $\frac{1}{4}$ cup of sugar. If Leo decides to bake 7 trays of brownies for the school bake sale, how many cups of sugar will he need in total?

12. Each student in Ms. Lopez's grade 4 class uses $\frac{2}{5}$ of a bottle of glue for their art project. If there are 6 students working at the main table, how many bottles of glue do they use combined?

13. A small puppy drinks $\frac{3}{8}$ of a bowl of water every morning. How many bowls of water will the puppy drink over a full week of 7 days?

14. Maya has 20 colored pencils in her pencil case. If $\frac{1}{5}$ of her pencils are bright green, how many green pencils does she have?

15. Tim needs to walk $\frac{3}{4}$ of a mile to reach the park. If he walks this exact same distance 5 days in a row, what is the total distance in miles he has walked?

✨ Keep up the amazing work! You are a fraction master! ✨